

Exercise 1: Data Profiling and Cleaning

Objective

In this exercise, we will explore the *House Prices* dataset from Kaggle and perform basic **data profiling** and **data cleaning**, focusing on identifying data types, missing values, and interpreting them using domain knowledge.

Step 1: Import Essential Libraries

Let us begin by importing the essential Python libraries required for this exercise.

- **NumPy**: Library for numerical computations
 - **Pandas**: Library for data acquisition and preparation
 - **Matplotlib**: Low-level library for data visualization
 - **Seaborn**: High-level library for statistical data visualization
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Step 2: Data Preparation

Dataset Description

We use the dataset from Kaggle's **House Prices: Advanced Regression Techniques** competition.

- **Source**: <https://www.kaggle.com/c/house-prices-advanced-regression-techniques>
- **File used**: train.csv

The dataset is already split into training and test sets. For this exercise, we will use **train.csv**.

Step 3: Quick Look at the Dataset

To understand the structure of the dataset, we inspect the first few rows using the `df.head()` function. This gives us an overview of: Column names, Sample values, Presence of missing values, Types of attributes (numeric vs categorical)

Step 4: Identify Data Types

We now examine the data types and non-null counts of each column using the `df.info()` function.

Observation

From the output, we observe:

- x variables of type int64
- y variables of type object (categorical)
- z variables of type float64

How did we get this count? Can we use `df.value_counts()` for finding these values?

Step 5: Check for Duplicate Records

Duplicate rows can bias analysis and models. We check for duplicates using:

```
df.duplicated().sum()
```

Step 6: Check for Missing Values in Numerical Columns which consists of int64 and float64 types.

What do you observe? Which ones have missing values and how many?

Step 7: Interpreting Missing Values

What do missing values indicate? How should we handle them?

- We should do Domain Interpretation: A missing value could mean: Data was not recorded OR The house does not have that attribute
- To validate this, we examine other related features.

Ex: One of the Missing Value is `GarageYrBlt`. To understand more of the same we need to investigate related features

We display rows where `GarageYrBlt` is missing along with: `GarageCars`, `GarageArea` and `GarageType`

Do the same for the other missing values.

What do you observe and what is your conclusion?

Key Learning Points: Not all missing values represent data errors, Some missing values carry semantic meaning, Related features should be examined before imputing, Domain knowledge improves data quality

Step 7: Detect zero values and see if that has an anomaly

We now examine the zero values. Some attributes can be zero which means that attribute is non-existent. But some attributes cannot be zero.

Step 8: Checking for outliers

- Using interquartile range, print the lower bound, upper bound and outlier data for “SalePrice”
 - Use Z score for checking outliers. You can keep the threshold to be 3.
 - Comment on when to use IQR and when to use Z score for checking outliers.
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Step 9: Checking relationship with “SalePrice” using correlation and plotting heatmap for the same

- Find the top 10 features which are highly correlated with sale price